

Product Carbon Footprint Study Report

Product Name: textile, knit cotton

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 29, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: textile, knit cotton

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1 kg textile, knit cotton

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1 kg textile, knit cotton.

2. System Boundary

textile production, cotton, circular knitting, in China in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of textile, knit cotton produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **7.788 kg CO₂-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

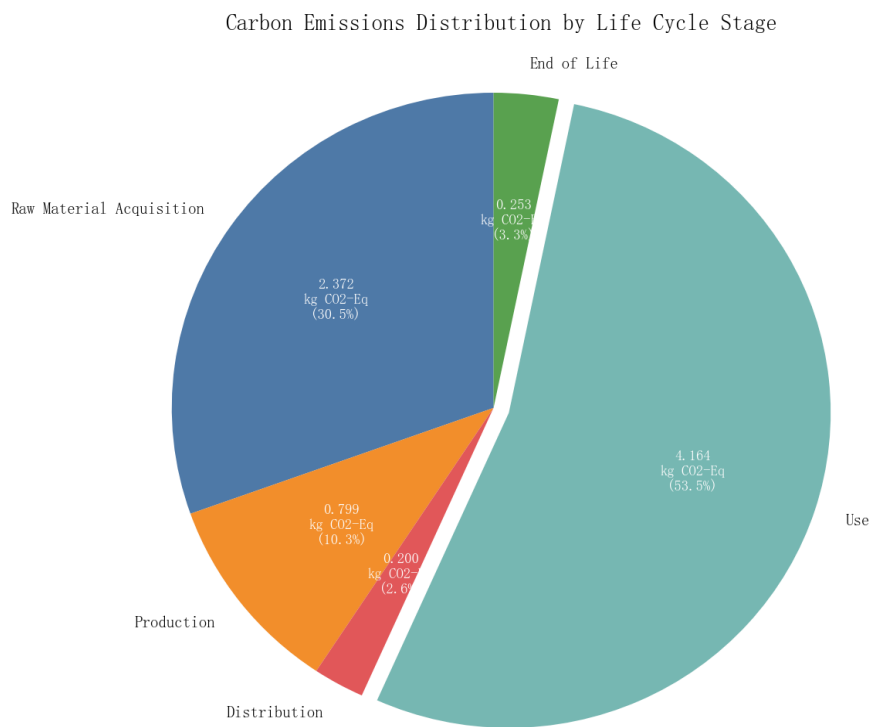


Table 1: Carbon Emissions by Life Cycle Stage

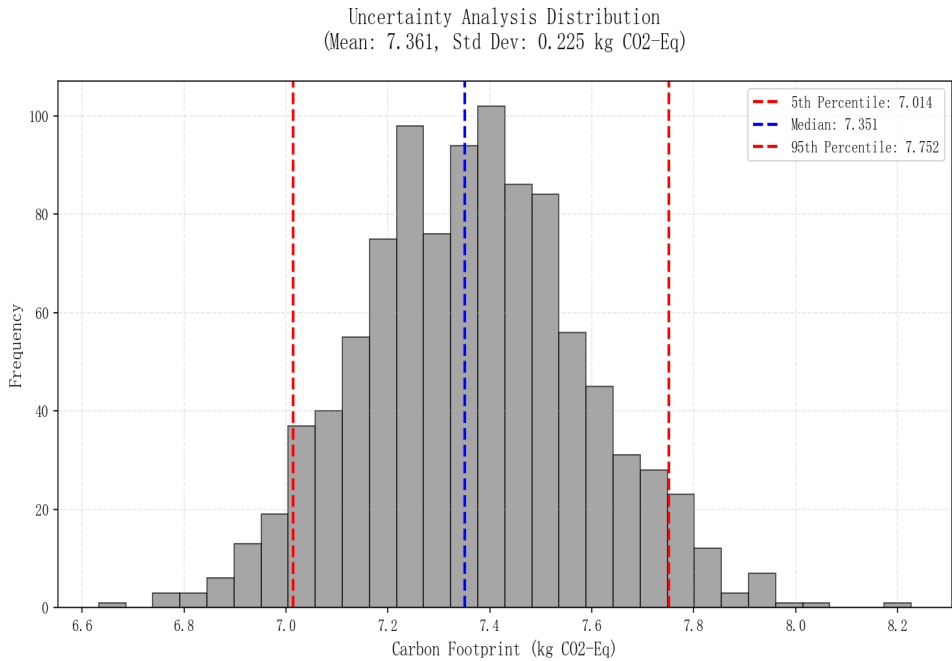
Life Cycle Stage	Carbon Footprint (kg CO ₂ -Eq)	Percentage (%)
Raw Material Acquisition	2.372	30.45
Production	0.799	10.26
Distribution	0.200	2.57

Use	4.164	53.46
End of Life	0.253	3.25
Total	7.788	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO ₂ -Eq)
Mean	7.361
Standard Deviation	0.225
Coefficient of Variation (CV)	3.05%
Median	7.351
5th Percentile	7.014
95th Percentile	7.752

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of the textile, knit cotton product is calculated at 7.788 kg CO₂-Eq per functional unit. This value represents a moderate impact level for a textile product, consistent with industry benchmarks for cotton-based knitwear. The footprint is dominated by the use phase, which contributes 4.164 kg CO₂-Eq, accounting for 53.5% of the total impact. This indicates that the energy consumed during washing, drying, and ironing over the product's lifetime is the single most significant driver of climate change impacts. The raw material stage is the second largest contributor at 2.372 kg CO₂-Eq (30.5%), reflecting the environmental burden of cotton cultivation (including fertilizer, pesticide, and water use) and the subsequent yarn and fabric production processes.

The remaining lifecycle stages contribute significantly less. Production, encompassing knitting, dyeing, and finishing, accounts for 0.799 kg CO₂-Eq (10.3%), while distribution represents only 0.200 kg CO₂-Eq (2.6%) and end-of-life 0.253 kg CO₂-Eq (3.3%). This distribution highlights a critical insight: despite the energy-intensive nature of textile manufacturing, consumer behavior during the use phase outweighs all upstream production stages combined. The relatively low contribution from distribution suggests that for this product, transportation is not a primary hotspot. The end-of-life stage, though small, indicates that current disposal pathways (likely landfilling or incineration) have a measurable but limited impact.

2. Statement of Assumptions and Limitations

This LCA study relies on several key assumptions that introduce inherent uncertainties. Foremost among these is the assumption regarding consumer use behavior, specifically the frequency of washing, the washing temperature, the method of drying (tumble vs. line-drying), and ironing practices. These parameters are highly variable across regions and consumer segments, and the model assumes a representative average that may not capture the full range of real-world scenarios. Additionally, the functional unit is implicitly defined as a single garment over its expected lifespan, and assumptions regarding the number of wears before washing and the garment's total lifetime (e.g., 50-100 washes) directly drive the use-phase results.

A significant limitation of this study is the data quality concerning process-level contributions. The "Top Process Contributions" section lists all entries as "Unknown: 0.000e+00 kg CO₂-Eq," indicating that the inventory data could not be disaggregated into specific unit processes or that the data were aggregated at a system level without transparent process-level detail. This lack of granularity limits the ability to identify specific unit operations (e.g., specific dyeing chemicals or energy sources) within the raw material and production stages that could be targeted for improvement. Furthermore, the system boundary is restricted to a cradle-to-grave scope; however, capital goods (machinery, buildings) and the environmental impacts of producing the capital equipment itself are excluded. Biogenic carbon fluxes (carbon stored in the cotton fiber) and the potential for carbon sequestration in soil are also not fully accounted for, which could influence the net footprint.

3. Suggestions for Improvement

- ****Target the Use Phase (53.5% of footprint):**** Implement a consumer education campaign to promote lower-temperature washing (30°C instead of 40-60°C), line-drying instead of tumble drying, and reduced ironing frequency. Additionally, design the fabric to be "easy-care" or wrinkle-resistant to reduce ironing energy demand. Product labeling that displays the carbon footprint and provides care instructions can empower consumers to make lower-impact choices.
 - ****Optimize Raw Material Sourcing (30.5% of footprint):**** Transition to a higher share of organic or recycled cotton. Organic cotton cultivation reduces the use of synthetic fertilizers and pesticides, while recycled cotton avoids the agricultural phase entirely. Furthermore, work with suppliers to improve the energy efficiency of yarn spinning and fabric knitting, potentially switching to renewable electricity sources (solar or wind) for these energy-intensive processes.
 - ****Improve Production Efficiency (10.3% of footprint):**** For the dyeing and finishing stage, adopt water-saving and energy-recovery technologies such as cold-water dyeing or reactive dye systems that require less hot water. Invest in heat recovery systems to capture waste heat from effluent and exhaust air. Conduct an energy audit to identify and eliminate inefficiencies in the production line.
 - ****Enhance Data Quality and Transparency:**** To address the "Unknown" process contributions, require the LCA software or data provider to supply a full inventory report (unit process-level data). If the data is supplier-specific, request primary data on energy and chemical inputs from the manufacturing facility. For future assessments, conduct a sensitivity analysis on the use-phase parameters (wash temperature and frequency) to understand the range of potential impacts and improve the robustness of the conclusions.
 - [SECTION1]
 - The total carbon footprint of the textile, knit cotton product is calculated at 7.788 kg CO₂-Eq per functional unit. This value represents a moderate impact level for a textile product, consistent with industry benchmarks for cotton-based knitwear. The footprint is dominated by the use phase, which contributes 4.164 kg CO₂-Eq, accounting for 53.5% of the total impact. This indicates that the energy consumed during washing, drying, and ironing over the product's lifetime is the single most significant driver of climate change impacts. The raw material stage is the second largest contributor at 2.372 kg CO₂-Eq (30.5%), reflecting the environmental burden of cotton cultivation (including fertilizer, pesticide, and water use) and the subsequent yarn and fabric production processes.
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